

Chapter 3 Problems

Scenario 1: The American Community Survey (ACS) is an ongoing survey that provides vital information on a yearly basis about the United States and its people. Information from the survey helps to determine where federal and state funds are distributed each year. Suppose that researchers were interested in the proportion of people who are employed that are looking for work elsewhere. Out of 83,922 employed respondents, 14,804 of them were actively looking for work. One such research question is as follows, "Are more than 15% of employed Americans looking for work?"

$$H_0: \pi = 0.15$$

$$H_A: \pi > 0.15$$

$$\hat{p} = \frac{14,804}{83,922} = 0.1764$$

Where π is the long run proportion of employed Americans who are looking for work.

- 1) Using the theory approach, calculate the standardized statistic.

$$Z = \frac{0.1764 - 0.15}{\sqrt{0.15(1-0.15)/83,922}} = 21.41$$

- 2) From your standardized statistic, do you reject or fail to reject the null hypothesis?

Reject / Fail to Reject)

- 3) Calculate the confidence interval for this sample. Remember that the SE formula for confidence intervals is different than the SE formula you used in Problem (1).

$$0.1764 \pm 2 \cdot \sqrt{\frac{0.1764(1-0.1764)}{83,922}} = (0.174, 0.179)$$

- 4) What is the interpretation of this confidence interval in the context of the scenario?

We are 95% certain that the long-run proportion of Americans who are employed and looking for work is between 0.174 and 0.179.

- 5) From your confidence interval, do you reject or fail to reject the null hypothesis? Why?

Reject, b/c 0.15 is outside of the CI

- 6) Write a conclusion for the research question using the context of the scenario.

We have evidence to conclude that the long-run proportion of employed Americans who are looking for work is greater than 0.15.

Scenario 2: The National Health Interview Survey (NHIS) is a survey collecting information on the health, health care access, and health behaviors of United States citizens. One question on the 2018 survey asked the respondents how many hours of sleep they usually get per night. The summary statistics for a similar study are below. You may assume that the data was not strongly skewed. The research question of interest is as follows, "Is the amount of sleep that Americans get different than the typical 8 hours."

Mean	Standard Deviation	Number of Respondents
7.8	2.3	200

$$H_0: \mu = 8$$

$$H_A: \mu \neq 8$$

- 1) Using the theory approach, calculate the standardized statistic.

$$A = \frac{7.8 - 8}{2.3 / \sqrt{200}} = -1.23$$

- 2) From your standardized statistic, do you reject or fail to reject the null hypothesis?

(Reject / Fail to Reject)

- 3) Calculate the confidence interval for this sample.

$$7.8 \pm 2 \cdot \frac{2.3}{\sqrt{200}} = (7.47, 8.13)$$

- 4) What is the interpretation of this confidence interval in the context of the scenario?

We are 95% confident that the long-run average number of hours of sleep that Americans get is between 7.47 and 8.13

- 5) From your confidence interval, do you reject or fail to reject the null hypothesis? Why?

FTR, b/c 8 is inside the CI

- 6) Write a conclusion for the research question using the context of the scenario.

We do not have evidence to conclude that the long-run average number of hours of sleep that Americans get is different than 8